## 10) Experiment to understand K-Means Clustering algorithm for a given data set.

Program:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

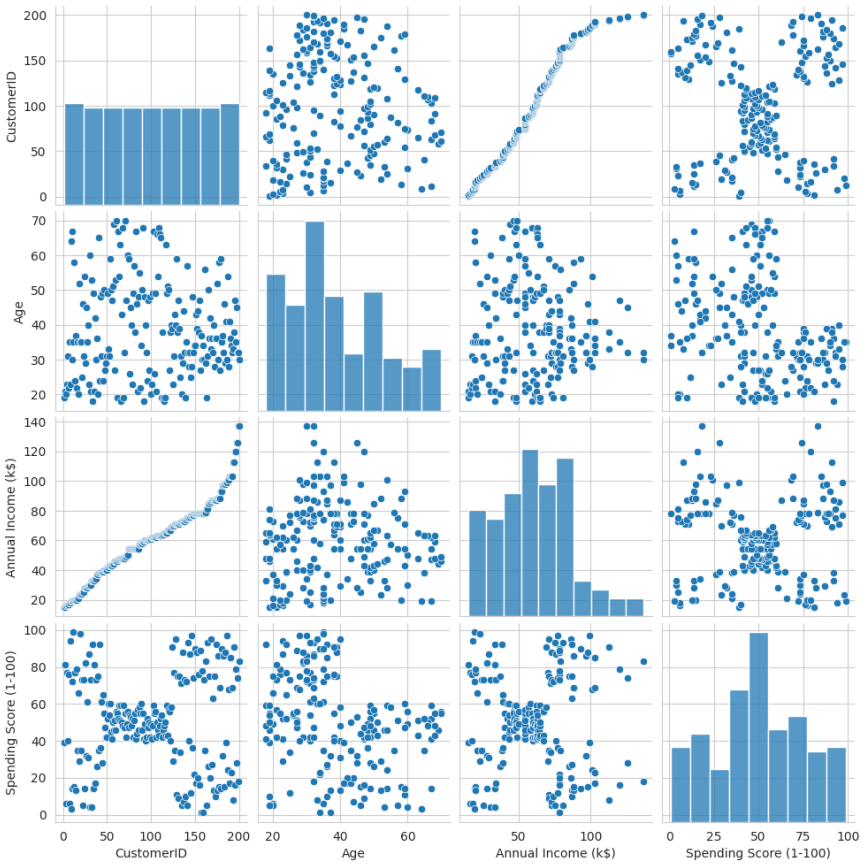

df = pd.read_csv('/Mall_Customers - Mall_Customers.xlsx - Mall_Customers - Mall_Customers.csv')
df.info()
display(df.head())
sns.pairplot(df)
plt.show()
features = df.iloc[:, [3, 4]].values
from sklearn.cluster import KMeans
model = KMeans(n_clusters=5, random_state=0, n_init=10)
model.fit(features)
Final = df.iloc[:, [3, 4]].copy()
Final['label'] = model.predict(features)
display(Final.head())
sns.set_style("whitegrid")
sns.FacetGrid(Final, hue="label", height=8) \
    .map(plt.scatter, "Annual Income (k$)", "Spending Score (1-100)") \
    .add_legend()
plt.show()
features_el = df.iloc[:, [2, 3, 4]].values
wcss = []
for i in range(1, 10):
```

```
    model = KMeans(n_clusters=i, random_state=0, n_init=10)
    model.fit(features_el)
    wcss.append(model.inertia_)
plt.plot(range(1, 10), wcss, marker='o')
plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.show()
```

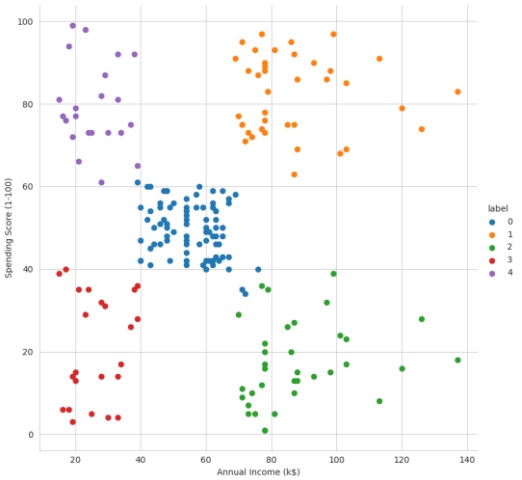
## Output:

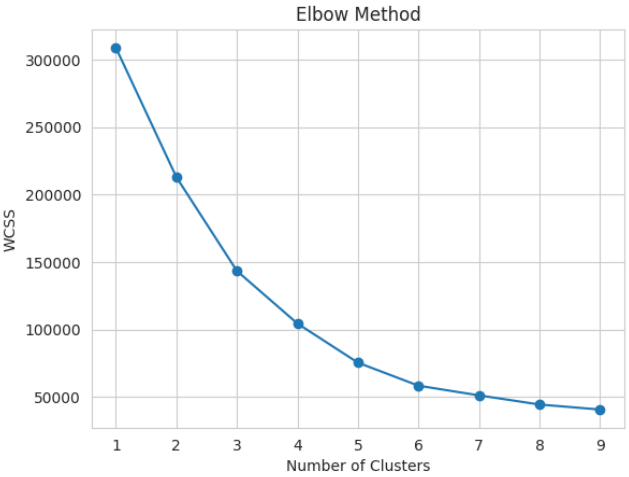
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

| | CustomerID | Gender | Age | Annual Income (k$) | Spending Score (1-100) |
|---|---|---|---|---|---|
| 0 | 1 | Male | 19 | 15 | 39 |
| 1 | 2 | Male | 21 | 15 | 81 |
| 2 | 3 | Female | 20 | 16 | 6 |
| 3 | 4 | Female | 23 | 16 | 77 |
| 4 | 5 | Female | 31 | 17 | 40 |




| | Annual Income (k$) | Spending Score (1-100) | label |
|---|---|---|---|
| 0 | 15 | 39 | 3 |
| 1 | 15 | 81 | 4 |
| 2 | 16 | 6 | 3 |
| 3 | 16 | 77 | 4 |
| 4 | 17 | 40 | 3 |



## Result:

Experiment is conducted to understand K-Means Clustering Algorithm of given Data set.